Overview of the Adaptive Immune Response

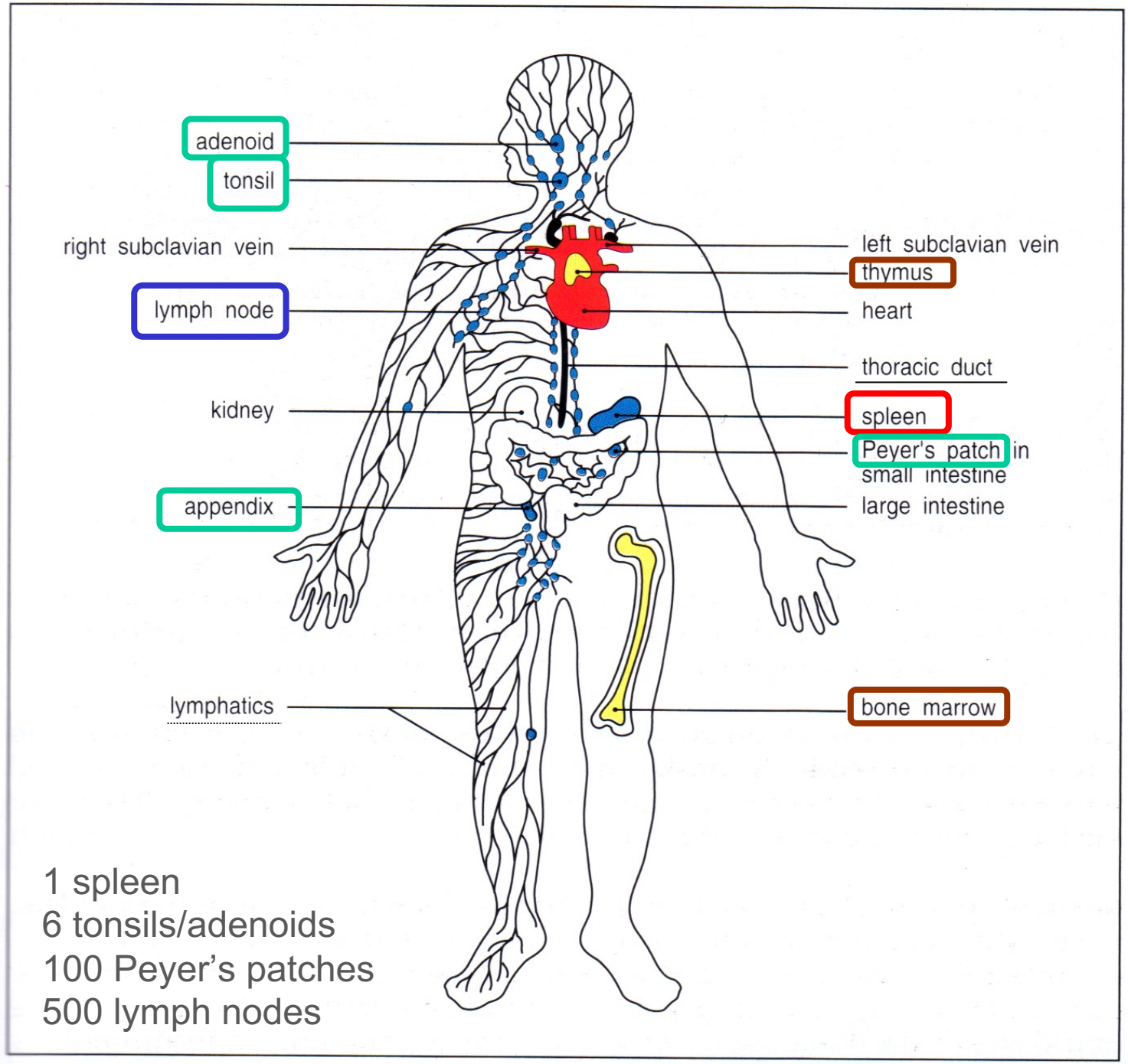
Lecturer: Jason Cyster

- Peripheral lymphoid tissues
- Trafficking (migration) from blood into tissue
- Antigen transport and presentation
- T cell–B cell interactions
- Activated B cells -> Plasma cells
- Effector T cells - traffic to sites of inflammation -> release cytokines, kill targets

Session Objectives

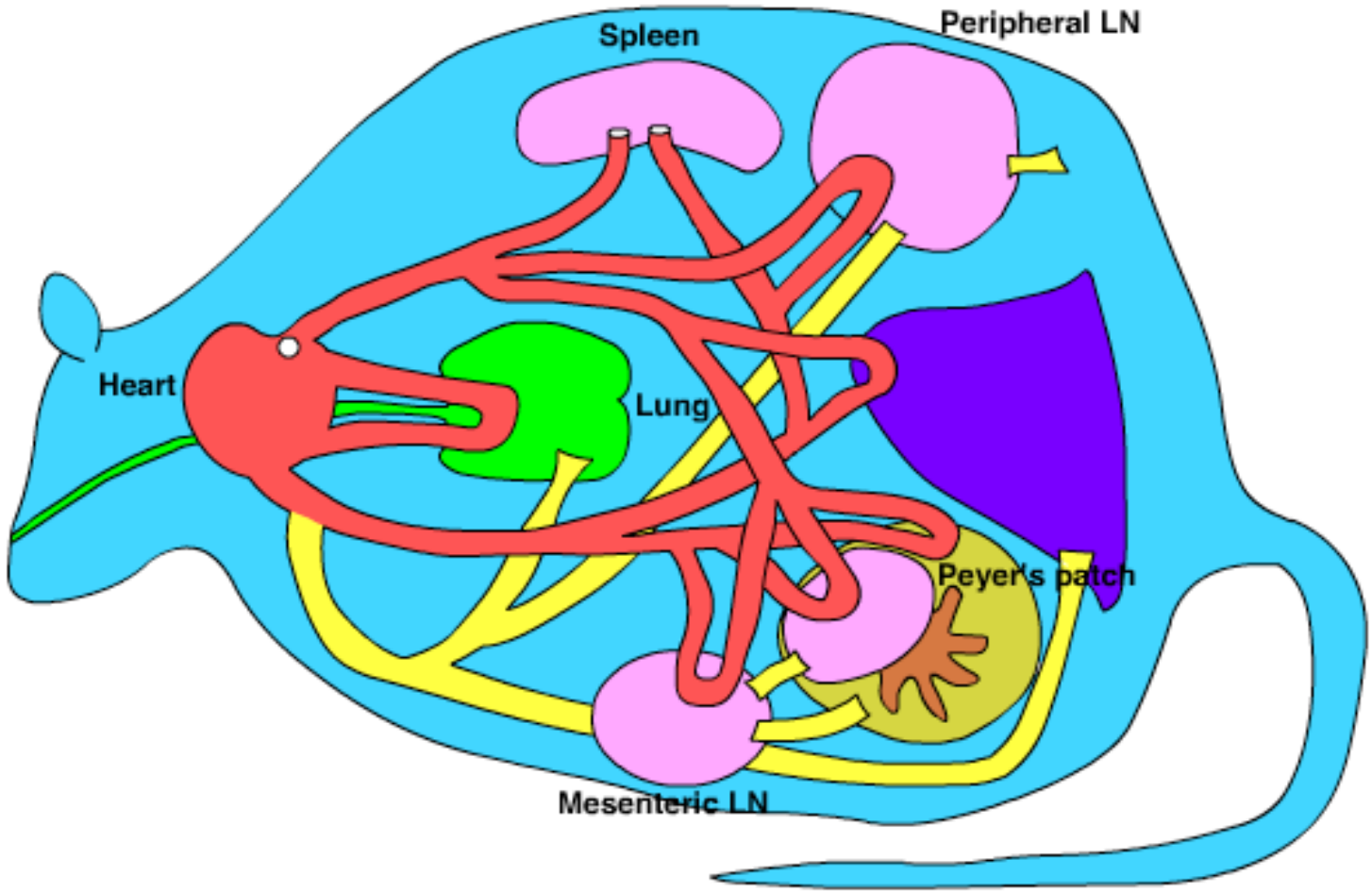
- Describe how lymphocytes migrate from the blood into lymph nodes including details about cell surface proteins and chemicals that participate in this process.
- Describe how and where naive lymphocytes encounter foreign antigens and what other cell types are involved in antigen presentation.
- Describe the important functions of dendritic cells in peripheral tissues and in lymphoid tissues.
- Describe how T cells and B cells become activated noting what cells and receptors are important for this to occur and where this typically happens.
- Describe the outcome of lymphocyte activation (i.e. how does activation impact the function of CD4 and CD8 T cells, and B cells).

Lymphoid Organs



1 spleen
6 tonsils/adenoids
100 Peyer's patches
500 lymph nodes

Recirculation of naïve lymphocytes



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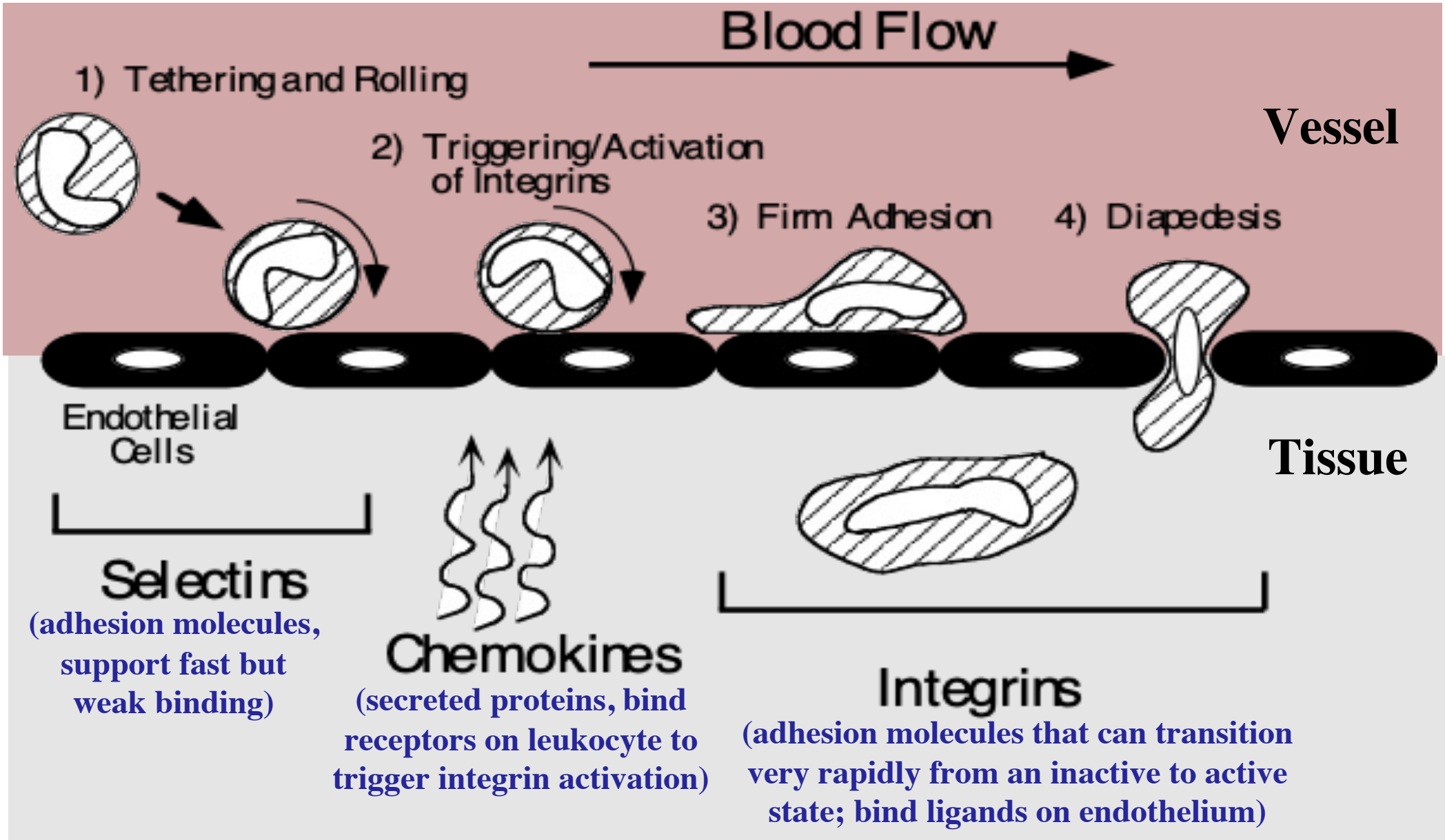
Animation provided by Marc Jenkins, Center for Immunology at the University of Minnesota

Immune function depends on ability of circulating leukocytes to move from blood into tissue

- Naïve lymphocytes (T and B cells) migrate from blood into secondary lymphoid organs (spleen, lymph nodes, Peyer's patches) to survey for antigen

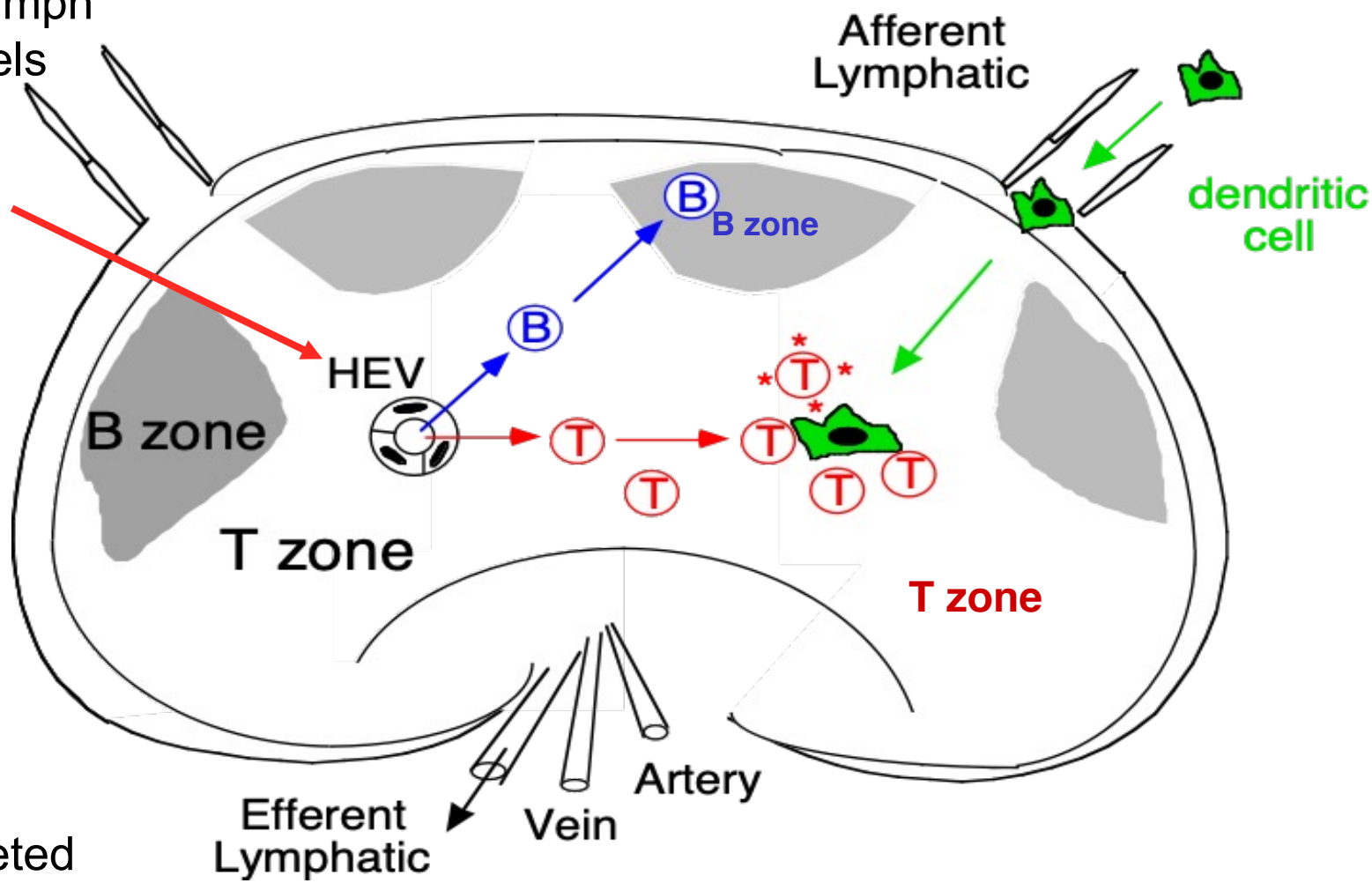
- Neutrophils, monocytes, and effector (activated) lymphocytes migrate across inflamed endothelium into infected tissue to fight the infection

The multistep model of leukocyte extravasation from blood into tissue



Schematic view of a lymph node

Lymphocytes enter lymph nodes by blood vessels known as HEV (high endothelial venules)



Chemokines are secreted proteins that promote directed cell migration - chemotaxis

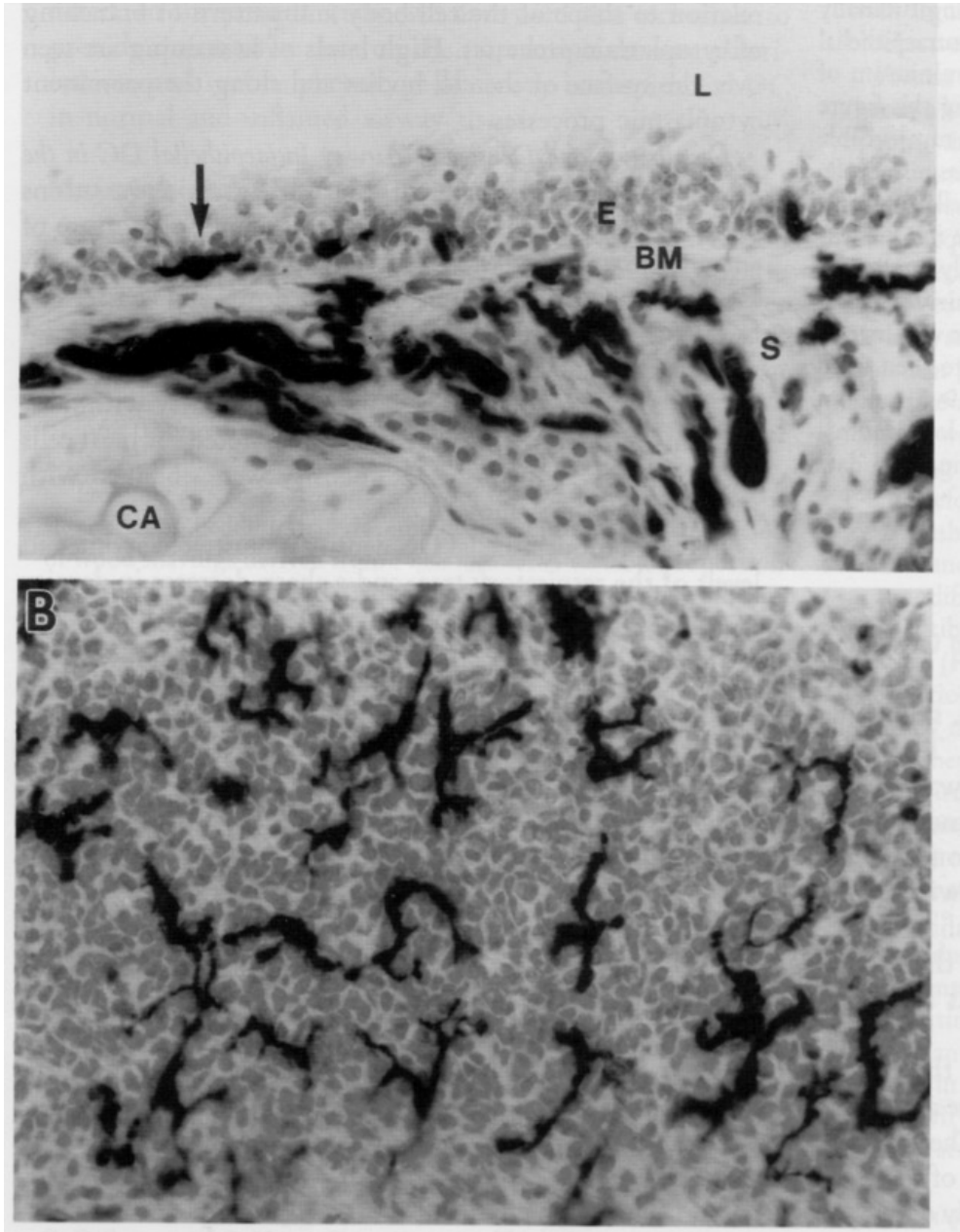
B zones (follicles) produce a B cell attracting chemokine

T zone produces T cell and DC attracting chemokines

Immature (sentinel) DCs in tracheal epithelium

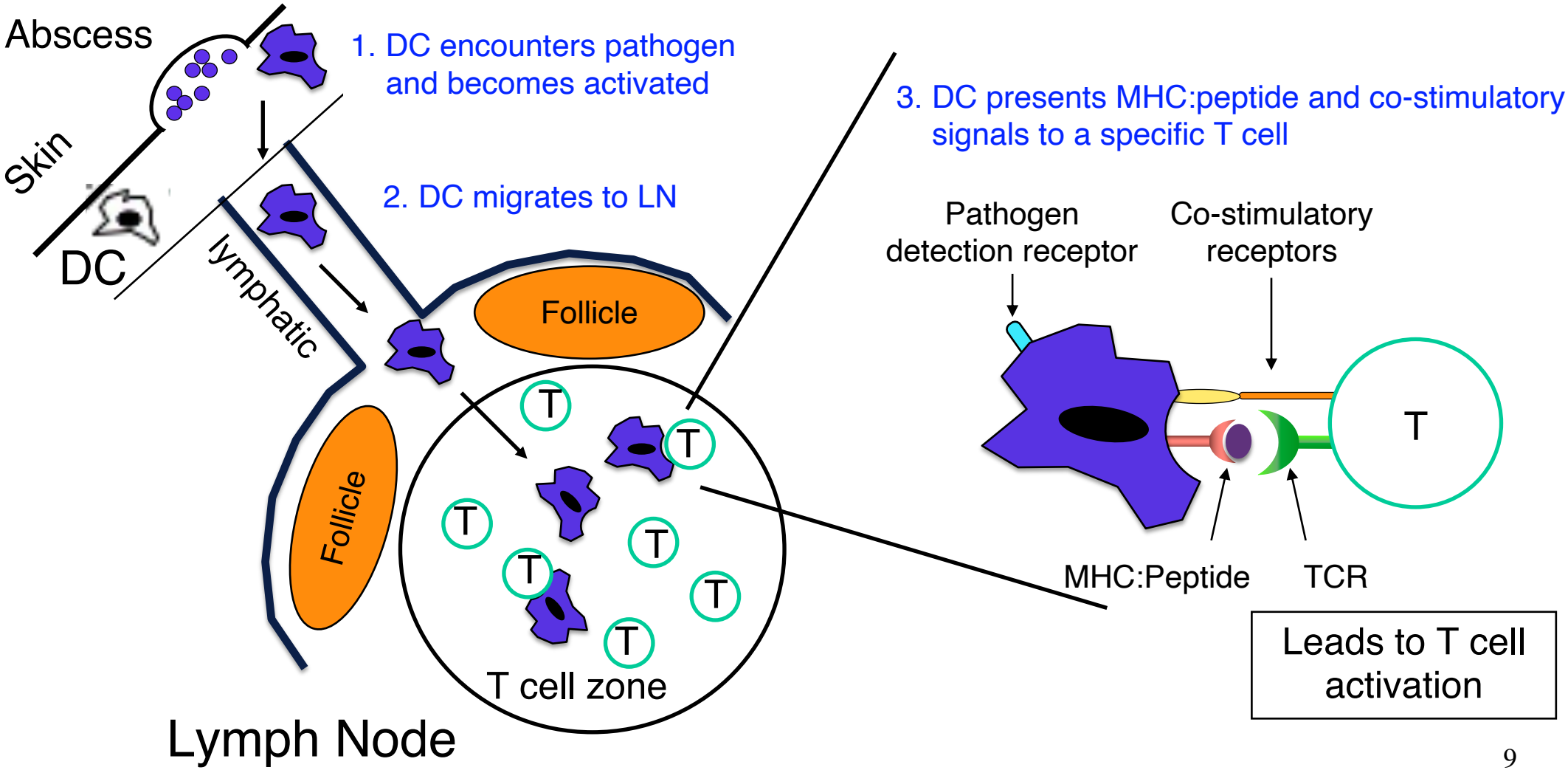
longitudinal section

tangential section



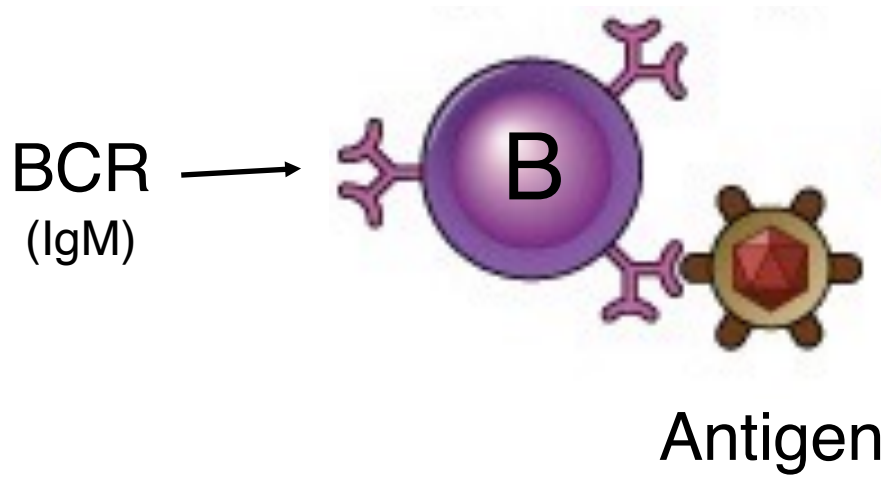
How do T cells encounter antigen?

Dendritic Cells (DCs) presenting processed peptide on MHC are key to initiating T cell-mediated immune responses in Lymph Nodes

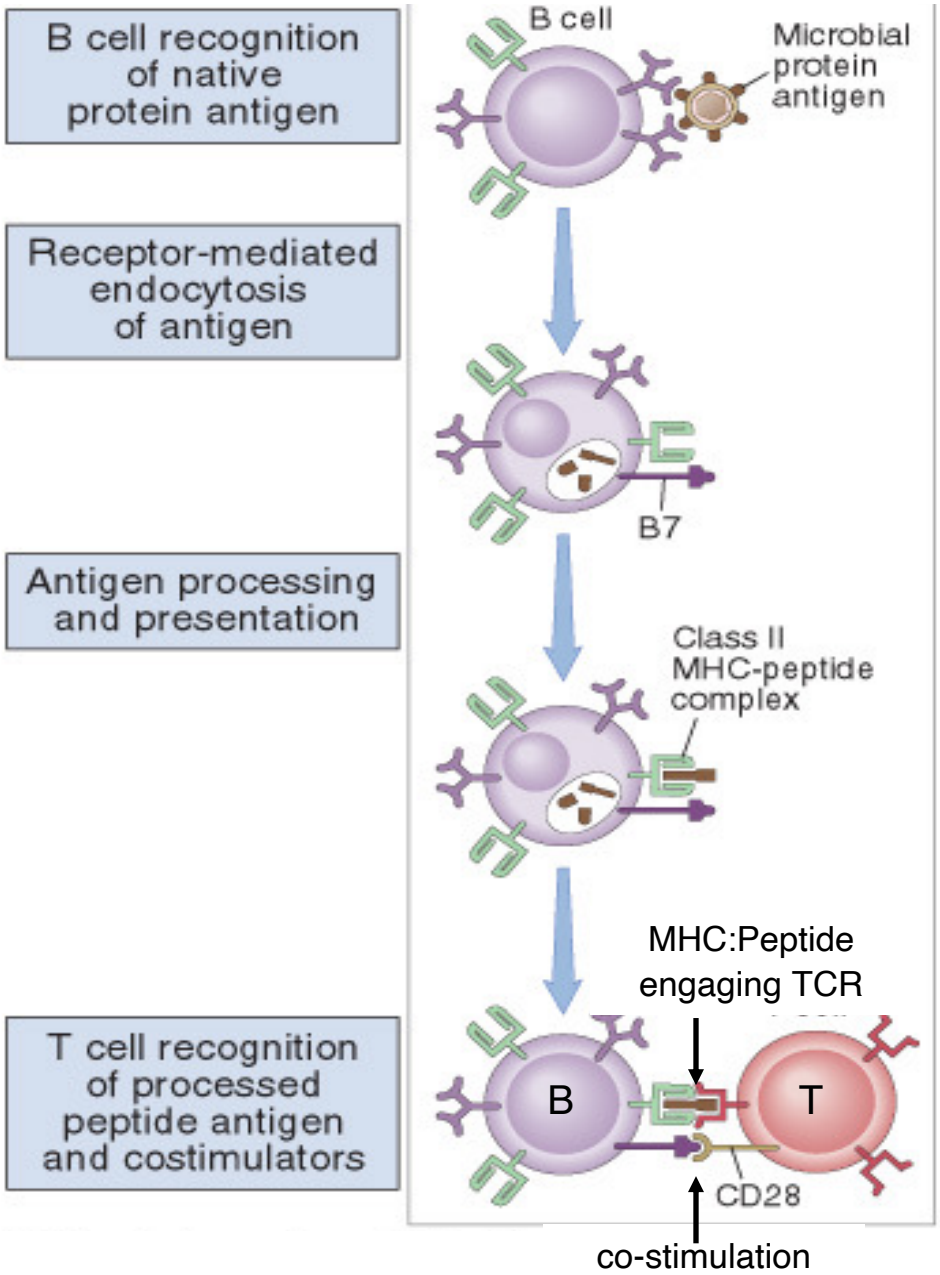


How do B cells encounter antigen?

- B cells bind <u>intact</u> antigen through their surface-localized B cell receptor (IgM)
- Can <u>directly bind</u> antigen that enters lymphoid tissues



Antigen presentation by B lymphocytes to helper T cells

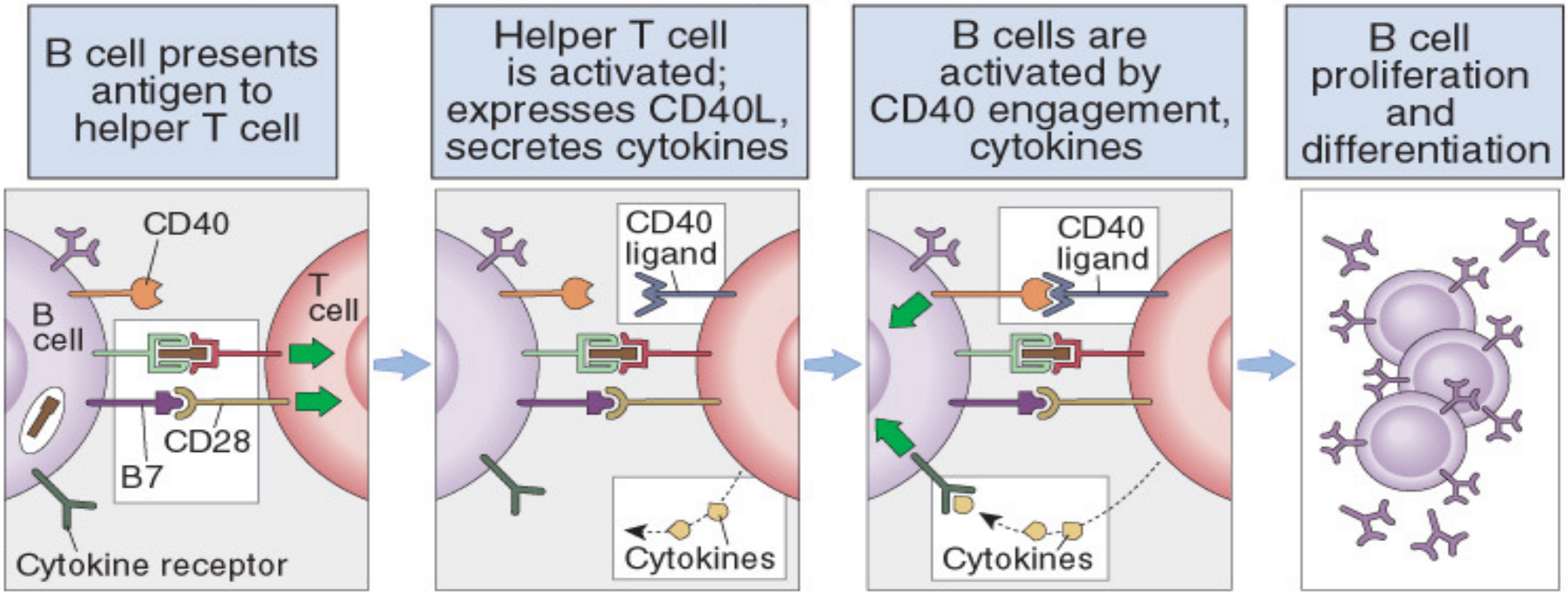


Naive B cell in a lymph node

The B cell acts as an APC, resulting in activation of the T helper cell. The activated T cell can then activate the B cell.

APC = antigen presenting cell

Mechanisms of helper T cell-mediated activation of B cells



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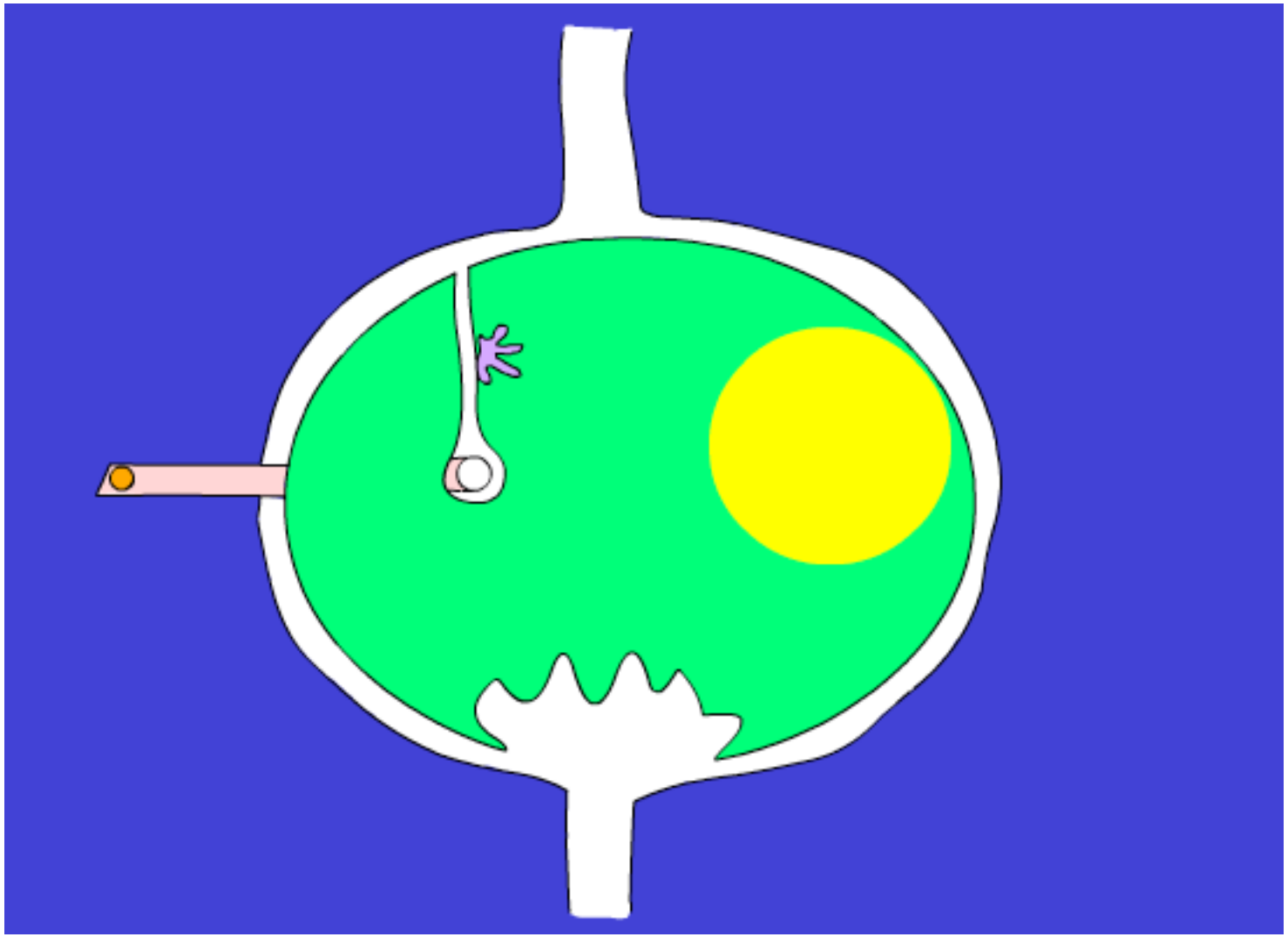
- Costimulatory molecule B7 is also known as CD86 – it engages CD28 on the T cell
- CD40L and cytokines (e.g., IL4) from T cell 'help' the B cell respond

Primary immune response in a lymph node



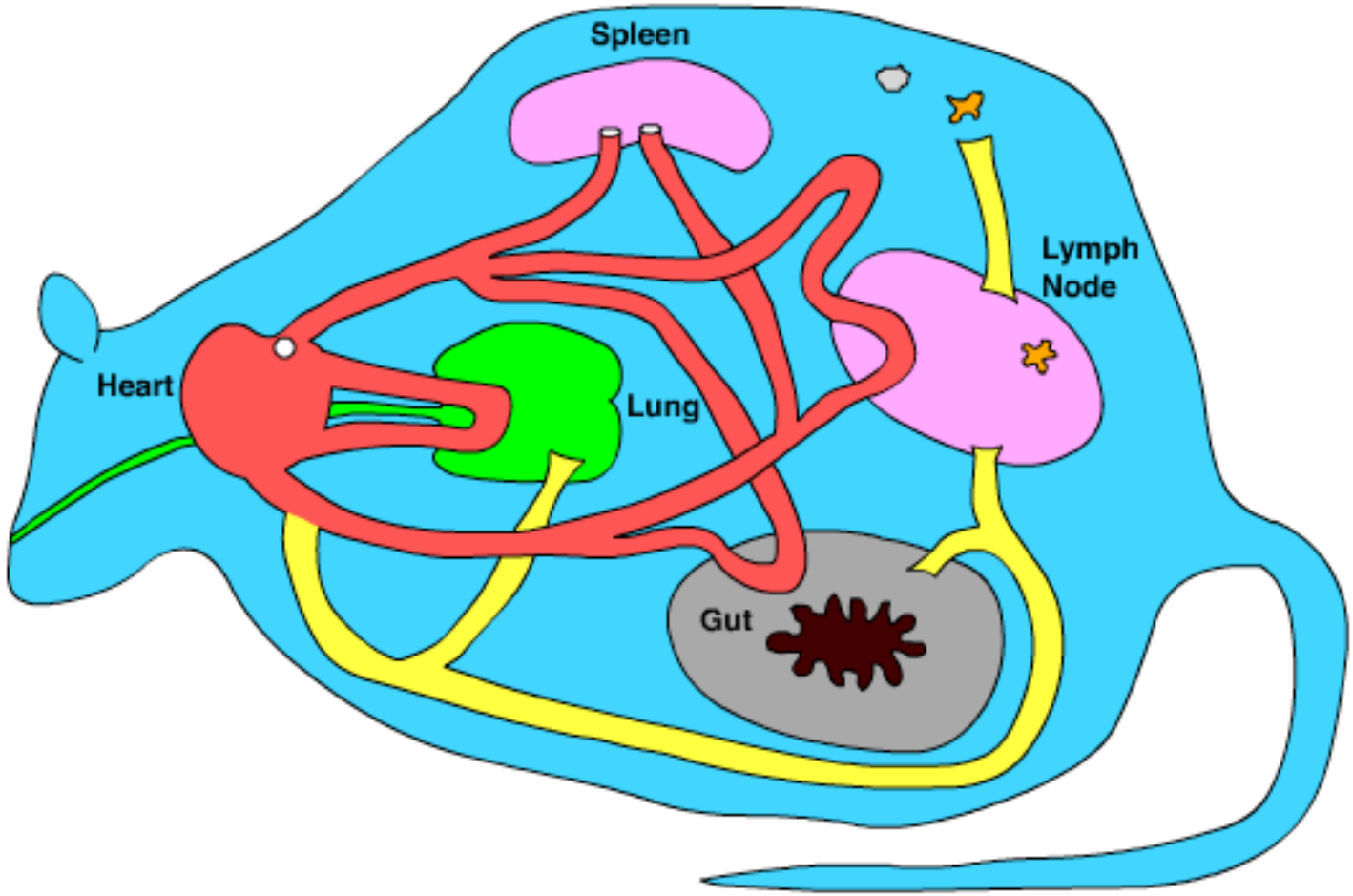
See also Jenkins et al., (2001)
Annu. Rev. Immunol. 19; 23

Animation provided by Marc Jenkins, Center for Immunology at the University of Minnesota

Effector T cell Trafficking

- **Activated T cells exit lymphoid tissue -> enter circulation**
  - Occurs after clonal expansion and taking on effector properties (e.g., ability to kill target cells or help macrophages kill ingested microbes)
  - Effector T cells have reduced ability to re-enter lymphoid tissue
- **Increased ability to enter inflamed tissue due to increased expression of:**
  - Receptors for inflammatory chemokines
  - Adhesion molecules that can interact with inflamed endothelium

Adaptive immune response to s.c. infection



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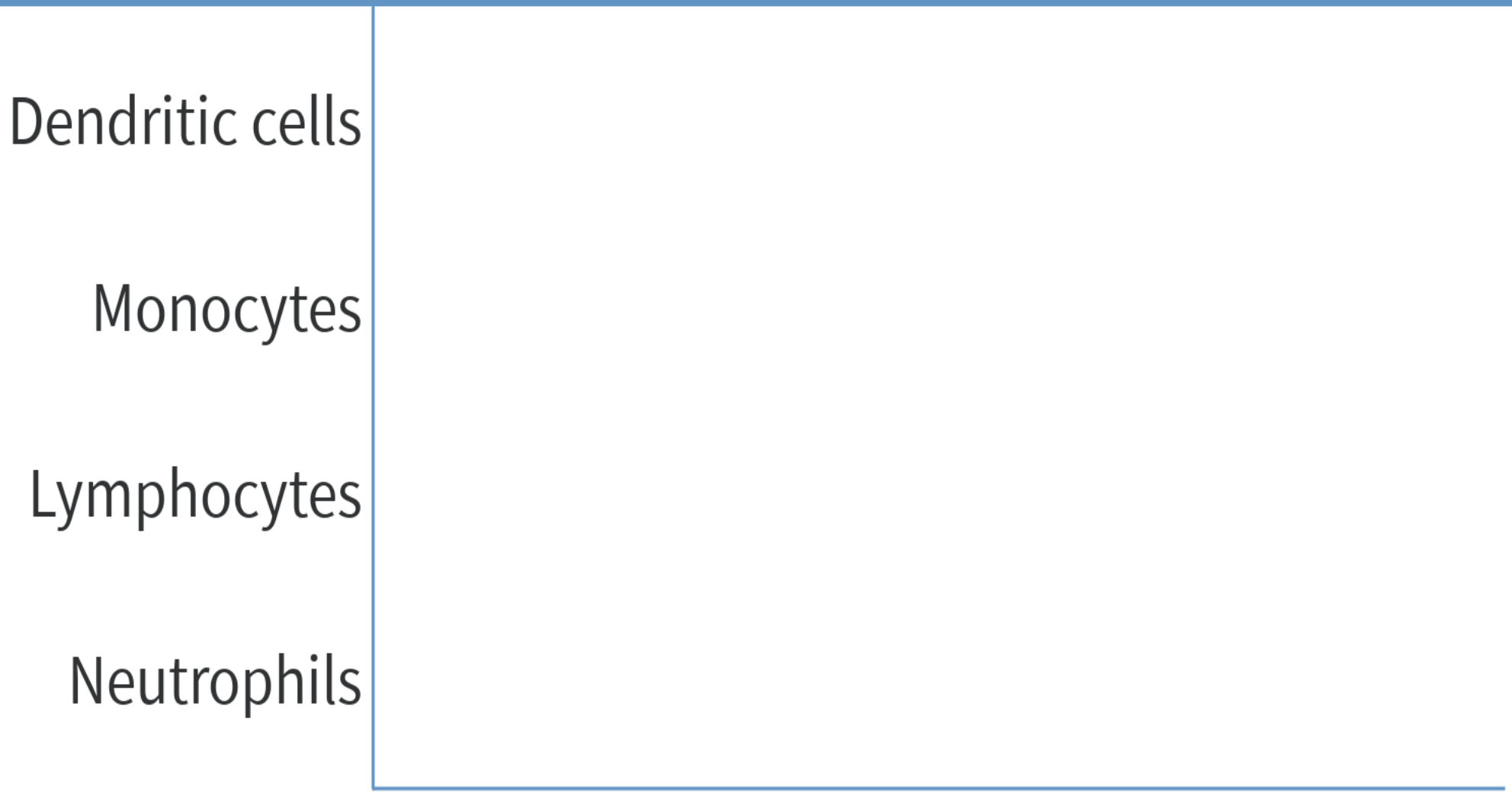
What cell types continually enter lymph nodes from blood?



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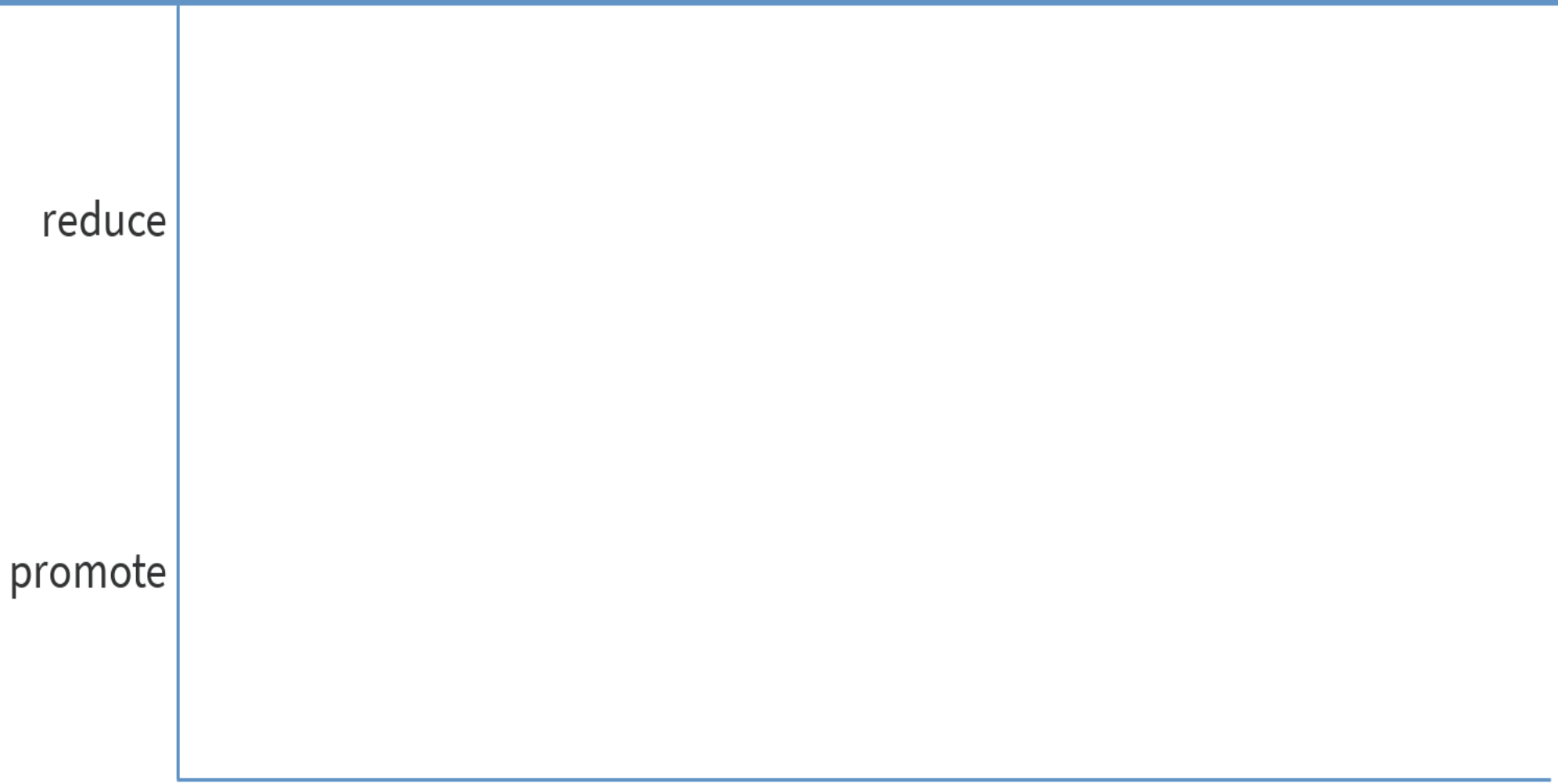
Do you think a blocking antibody against an adhesion molecule (e.g. an integrin) on the surface of a T cell would reduce or promote inflammatory disease?



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Which lymphoid tissue is most likely to support the response to an inhaled antigen?

Spleen	**A**
Lymph node	**B**
Peyer's patch	**C**
Tonsil / adenoids	**D**
B and D	**E**
A and C	**F**

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Text **JASONCYSTER262** to **37607** once to join, then **A or B**

If you lack dendritic cells do you expect to have a reduced T cell-dependent antibody response?

Yes **A**

No **B**

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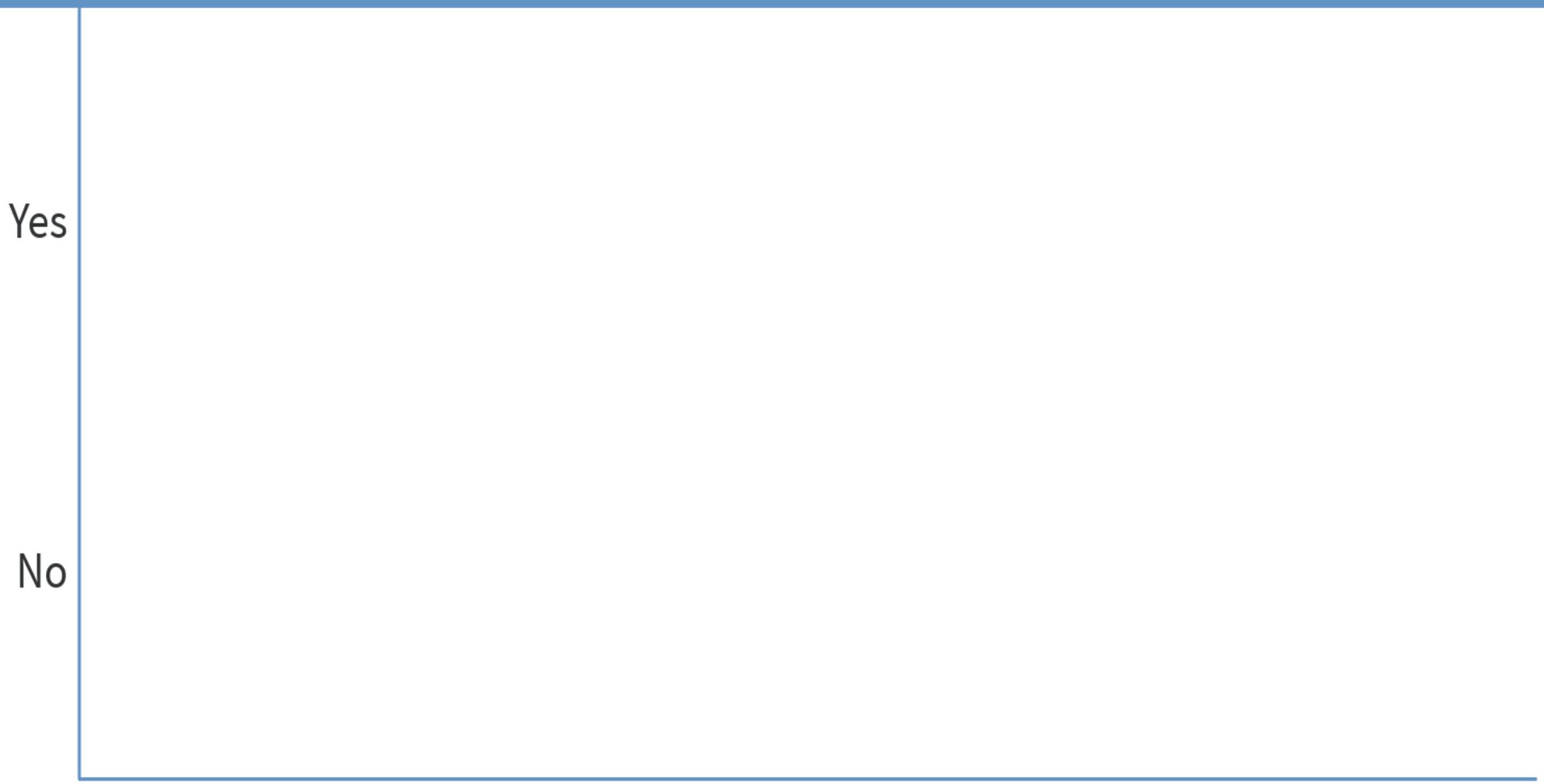
For a T-dependent antibody response do the B cell and T cell have to see the exact same part of the antigen?



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